EXERCISE 4:

**DEVELOP A CLIENT SIDE APPLICATION USING REACT API A) FORM DESIGN AND VALIDATION B)APPLICATION USING REACT LIST c) APPPLICATION USING SaaS**

**Aim**

To develop a client-side application using React for an Internship Tracker that includes:

- Form design with validation to collect internship details
- Dynamic list rendering using React
- Styling using Sass for better UI design

**Step 1: Setup React Environment**

1. Install Node.js
2. Create a React app using:

```
npx create-react-app internship-tracker
cd internship-tracker
npm start
```

**Step 2: Form Design & Validation**

1. Create a form component to collect:
    - Company Name
    - Role
    - Duration
    - Status
2. Use useState to manage form data
3. Validate inputs:
    - No empty fields
    - Minimum character length
4. Show error messages dynamically

**Step 3: React List Application**

1. Store internship entries in state (array)
2. On form submission:

    - Add new internship object to list
3. Display internships using .map()
4. Allow deletion of entries

**Step 4: Styling using Sass**

1. Install Sass:

npm install sass

2. Create .scss files
3. Use variables, nesting, and styling for UI enhancement

SOURCE CODE:

**App.js:**

```
import React, { useState } from "react";
import "./App.scss";
function App() {
  const [form, setForm] = useState({
    company: "",
    role: "",
    duration: "",
    status: ""
  });
  const [errors, setErrors] = useState({});
  const [internships, setInternships] = useState([]);
  const validate = () => {
    let tempErrors = {};
    if (!form.company) tempErrors.company = "Company required";
    if (!form.role) tempErrors.role = "Role required";
    if (!form.duration) tempErrors.duration = "Duration required";
    if (!form.status) tempErrors.status = "Status required";
    setErrors(tempErrors);
```

```
  return Object.keys(tempErrors).length === 0;
};
const handleChange = (e) => {
  setForm({ ...form, [e.target.name]: e.target.value });
};
const handleSubmit = (e) => {
  e.preventDefault();
  if (validate()) {
    setInternships([...internships, form]);
    setForm({ company: "", role: "", duration: "", status: "" });
  }
};
const deleteItem = (index) => {
  const updated = internships.filter((_, i) => i !== index);
  setInternships(updated);
};
return (
  <div className="container">
    <h1>Internship Tracker</h1>

    <form onSubmit={handleSubmit}>
      <input
        name="company"
        placeholder="Company Name"
        value={form.company}
        onChange={handleChange}
      />
      <p className="error">{errors.company}</p>
```

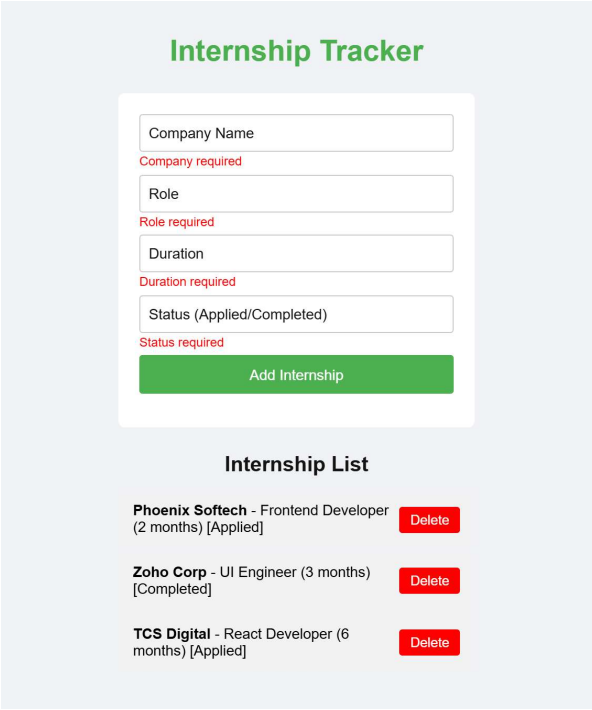
```
  <input
    name="role"
    placeholder="Role"
    value={form.role}
    onChange={handleChange}
  />
  <p className="error">{errors.role}</p>
  <input
    name="duration"
    placeholder="Duration"
    value={form.duration}
    onChange={handleChange}
  />
  <p className="error">{errors.duration}</p>

  <input
    name="status"
    placeholder="Status (Applied/Completed)"
    value={form.status}
    onChange={handleChange}
  />
  <p className="error">{errors.status}</p>
  <button type="submit">Add Internship</button>
</form>
<h2>Internship List</h2>
<ul>
  {internships.map((item, index) => (
    <li key={index}>
```

```
          <strong>{item.company}</strong> - {item.role} ({item.duration}) [{item.status}]
          <button onClick={() => deleteItem(index)}>Delete</button>
        </li>
      ))}
    </ul>
  </div>
 );
}
export default App;
$primary-color: #4CAF50;
$error-color: red;
.container {
 width: 60%;
 margin: auto;
 text-align: center;
 h1 {
  color: $primary-color;
 }
 form {
  margin-bottom: 20px;
  input {
   display: block;
   width: 100%;
   padding: 8px;
   margin: 5px 0;
  }
  button {
   background: $primary-color;
```

```
      color: white;
      padding: 10px;
      border: none;
      cursor: pointer;
    }
  }
  .error {
    color: $error-color;
    font-size: 12px;
  }
  ul {
    list-style: none;
    padding: 0;
    li {
      background: #f2f2f2;
      margin: 5px;
      padding: 10px;
      button {
        margin-left: 10px;
        background: red;
        color: white;
        border: none;
        padding: 5px;
      }
    }
  }
}
```

**OUTPUT:**




**Result**

The Internship Tracker application was successfully developed using React.